Quantum sensor enables GPS-free maritime navigation

Q-CTRL has demonstrated the world's first GPS-free quantum gravimetric navigation of a maritime vessel during field trials in Australia's Coral Sea. Its Ironstone Opal system combines a quantum gravimeter with inertial navigation, continuously matching Earth's gravity variations against existing maps to correct positioning drift. AI-powered software stabilises the sensor against movement and environmental disturbances, allowing autonomous operation without specialised installation or additional motion-control infrastructure. Achieving accuracy of about one nautical mile (1852m), the technology offers a passive alternative for resilient maritime navigation where GNSS signals are unavailable, jammed, or disrupted.



Q-CTRL achieves world's first GPS-free quantum gravimetric navigation demonstration in maritime field trial (Credit: https://q-ctrl.com/blog)

Coating method prepares TGVs for AI chips

TRUMPF has developed an industrial HiPIMS process designed to uniformly coat microscopic through-glass vias (TGVs) within complex glass substrates, addressing a key manufacturing challenge in next-generation AI chip packaging. The process generates highly ionised plasma and uses electric and magnetic fields to drive conductive material into narrow, deep glass structures. Designed to improve process stability and manufacturing yield for advanced packaging, the infrastructure targets high-performance chips utilised across artificial intelligence, data centres, and high-performance computers where dense interconnects and faster signal transmission are increasingly important. The configuration works alongside ultrashort-pulse lasers for via production and plasma power supplies for coating and etching.



Testing the parameters of plasma generators at TRUMPF's research and development laboratories in Zielonka, Poland (Credit: TRUMPF)

Cobalt silicide wiring beats copper tenfold at 20 nanometres

Single-crystalline cobalt silicide (CoSi) has demonstrated unusually low resistance at nanoscale dimensions, emerging as a potential alternative to copper for advanced chip interconnects. At a thickness of 20nm, CoSi achieved a resistivity of 0.72 microohm-centimetres, which is about a tenth that of copper. Developed by researchers at The Hong Kong Polytechnic University, the material becomes more conductive as it thins, due to highly conductive surface states. CoSi also supports current densities up to 100 million A/cm² and temperatures up to 450°C. Integrated into a 16nm node silicon ring oscillator, it matched conventional interconnect performance, indicating strong potential for future high-performance chips.

Seven-dollar LED case helps phones find hidden cameras

Researchers from KAIST, the National University of Singapore, and Singapore Management University have developed a low-cost smartphone attachment that detects concealed cameras by analysing lens reflections. The LED-based device illuminates a scene from multiple angles, while a deep-learning model identifies the distinctive glint produced by curved camera lenses. Tested on 30 real-world objects, the system achieved about 94 per cent detection accuracy, with each inspection taking less than five seconds. Priced under $7, this advance could make hidden-camera detection more accessible for travellers, hotel guests, and other users without specialised security equipment.



Each inspection takes less than five seconds, with a deep-learning model analysing how light reflects off camera lenses versus ordinary objects (Credit: KAIST)

Micro-sized photonic chip enables higher-capacity 6G networks

Researchers at Loughborough University's Emergent Photonics Research Centre, working with international collaborators, have demonstrated a rice-grain-sized photonic chip that produces a precisely organised 'rainbow' of light frequencies, known as a microcomb. By combining a chip-based microresonator with an optical fibre loop, the system generates stable, controllable optical frequencies that can be converted into multiple high-frequency millimetre-wave signals. This breakthrough preserves precision and stability, opening the door to higher-capacity 6G networks and supporting applications in radar, spectroscopy, navigation, positioning, and quantum technologies requiring ultra-accurate timing.